

Avijeet Shil

Ph.D. Candidate in Computer Science and Engineering
University of North Texas

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[avijeetas.github.io](https://github.com/avijeetas) | Google Scholar - <https://scholar.google.com/citations?user=NuIbgCAAAAAAJ>

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Education

University of North Texas

Denton, TX

Ph.D. in Computer Science & Engineering (GPA: 4.00 / 4.00)

Aug 2023 – May 2027 (Expected)

- Research Interests: Biomedical Natural Language Processing, Retrieval-Augmented Generation (RAG), Learning-to-Rank, Knowledge Graphs, Large Language Models, Scientific Information Retrieval
- Advisor: Dr. Wei Jin

University of North Texas

Denton, TX

M.Sc. in Computer Science (GPA: 4.00 / 4.00)

Aug 2023 – July 2026

Chittagong University of Engineering & Technology

Bangladesh

B.Sc. in Computer Science & Engineering (GPA: 3.56 / 4.00)

Aug 2021

Publications

[U] Under Review [A] Accepted [P] Published

- [1] **Shil, A.**, & Jin, W. (2026). *HyBRE: A Hybrid Ranking Retrieval Framework for Biomedical Document-Level Relation Extraction*. [A]
- [2] **Shil, A.**, & Jin, W. (2026). *Structural Signals Matter: Gated Multi-Signal Learning-to-Rank for Biomedical Relation Extraction*. [U]
- [3] Florescu, C., **Shil, A.**, & Jin, W. (2026). *Keyphrase Extraction using Interpretable Network Representation Learning*. [U]
- [4] **Shil, A.**, & Jin, W. (2025). *SQUIRE+LLM: A LLM-assisted ESA-based Unsupervised Semantic Query Interpretation and Ranking Engine*. IEEE/WIC WI-IAT 2025. [A]
- [5] **Shil, A.**, & Jin, W. (2025). *ESAQueryRank: Ranking Query Interpretations for Document Retrieval Using Explicit Semantic Analysis*. In Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP 2025), pp. 1148–1152, Varna, Bulgaria. INCOMA Ltd. aclanthology.org/2025.ranlp-1.132. [P]
- [6] Florescu, C., **Shil, A.**, & Jin, W. (2024). *A New Learning-to-Rank Framework for Keyphrase Extraction Using Multi-scale Ratings and Feature Fusion*. APWeb-WAIM 2024, pp. 63–79. Springer Nature. [P]
- [7] Zubair, M., Iqbal, M. A., **Shil, A.**, et al. (2022). *An Improved K-Means Clustering Algorithm Towards an Efficient Data-Driven Modeling*. Annals of Data Science. [P]
- [8] Zubair, M., Iqbal, M., **Shil, A.**, et al. (2021). *An Efficient K-Means Clustering Algorithm for Analysing COVID-19*. HIS 2020, Springer AISC vol. 1375. [P]

Research Experience

Research Assistant

Denton, TX

University of North Texas — Texas OneGulf (PIs: Ashley Ross, Laura Siebeneck, Tristan Wu) *May 2026 – Dec 2026*

- Developed and maintained **DynaSearch**, a research platform for the Texas OneGulf project *Risk Communication in an Age of Misinformation: Emergency Manager and Household Perspectives*, supporting studies on hurricane risk communication, emergency messaging, and misinformation to improve disaster preparedness and public response.
- Designed and implemented experimental web interfaces, including randomized content presentation and multimedia display modules, to support controlled behavioral studies on public responses to hurricane advisories.
- Managed CI/CD pipelines and production deployment, organized hurricane advisory and misinformation datasets, and supported pilot studies and data analysis for interdisciplinary research in emergency management and disaster science.

Summer Camp Graduate Research Assistant — AI & Embedded Systems Instructor **Denton, TX**
University of North Texas *May 2025 – Jul 2025*

- Developed and delivered hands-on AI projects using MicroPython and Kendryte K210-based development boards.
- Instructed students on foundational to advanced concepts in artificial intelligence, computer vision, and embedded programming.
- Led practical demonstrations of real-world AI applications, including image recognition, object detection, and face tracking.
- Created tutorials and step-by-step guides for beginner and intermediate learners (170 + *students*).

Graduate Researcher **Denton, TX**
University of North Texas — Text and Web Mining Lab (Advisor: Dr. Wei Jin) *Aug 2023 – Present*

- Built **GatedRanker**, a multi-signal LTR pipeline fusing semantic relevance, structural dependency paths, and co-occurrence features; achieved $F1 = 0.4876$ on CDR (~19% relative gain over semantic-only baseline).
- Designed chain-of-thought (CoT) inference pipeline for LLM-based biomedical relation classification across CDR, GDA, and ChemDisGene datasets using LLaMA 3.1 (ChatOllama/LangChain).
- Developed cache-based ETL pipelines processing 500K+ biomedical documents/day, improving throughput by 3× and reducing latency by 70%.
- Applied GraphRAG to biomedical corpora, improving information relevance by 15% and model interpretability.
- Built **ESAQueryRank** semantic query ranking engine, achieving Mean Reciprocal Rank (MRR) 480% and query response latency 80%.

Teaching Experience

Teaching Assistant L2 **Denton, TX**
University of North Texas *Aug 2023 – Present*

- Courses: Software Development Capstone I & II, Software Engineering.
- Mentored 80+ capstone teams on projects including AI accessibility tools, VR CS education, graph sonification, and a research participant recruitment portal.
- Mentored them mastering different programming languages (Java, Python, C++) by providing useful suggestions, and materials.

Industry Experience

Software Engineer I **Dhaka, Bangladesh**
Vivasoft Limited (bKash — Bangladesh's largest Fintech platform) *Nov 2022 – Aug 2023*

- Designed custom API rate-limiter improving TPS by 50+ per microservice using Spring Boot; migrated monoliths to microservices improving scalability by 40%.
- Built Webhook/SMS configuration system using Reactive Java (WebFlux), AWS SNS/SQS.
- Developed unified backend API serving 3 fintech platforms across 9 financial system types.

Software Engineer — E-Governance **Dhaka, Bangladesh**
REVE Systems *Jun 2021 – Oct 2022*

- Designed NLP-based duplicate question detection API: 98% accuracy, 150ms inference, over a 1,000+ question bank.
- Built automated report management system reducing creation time by 85%; processed 500+ daily reports at 99.99% accuracy.
- Implemented role-based notification and approval workflows covering 13 distinct business task types.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, GoLang

AI & NLP: Large Language Model, Learning-to-Rank, Retrieval Augmented Generation (RAG), GraphRAG, Knowledge Graphs, GNNs, Biomedical NLP, Relation Extraction

ML and DL frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, Scikit-learn, Keras, FastAPI, matplotlib

Infrastructure: Docker, Kubernetes, AWS, GCP, Azure, FAISS, Redis, PostgreSQL, MongoDB, Hadoop

Other: L^AT_EX, Git, Reactive Programming, Algorithm Design, Distributed Systems

Honors & Awards

TGS Student Travel Grant (\$500) — University of North Texas	Jan 2026
TGS Student Travel Grant (\$1,250) — University of North Texas	Aug 2025
Porter-Evans Scholarship (\$8,000+) — University of North Texas	Jun 2025
Toulouse Graduate School Tuition Benefit (\$4,140 per semester)— University of North Texas	Aug 2023 – Present
CENG/DT Award (85%) — University of North Texas	Aug 2023 – Present
Merit-Based Scholarship — Higher Secondary Certificate (Top 150, Education Board Bangladesh)	2016 – 2021
Merit-Based Scholarship — Secondary School Certificate (Top 250, Education Board Bangladesh)	2013 – 2015
Golang Coding Bootcamp — Top 10 / 800+ participants	May 2021
ACM ICPC Dhaka Regional 2018 — Top 5 CUET University Teams	Dec 2018
EDU Engineering Day Programming Contest 2018 — 13th position	Sept 2018
CUET Inter-University Programming Contest 2017 — 8th overall, Top 3 CUET Team	Nov 2017

Professional Service & Leadership

Capstone Mentor (I & II) — Mentored senior design projects on AI, Machine Learning, Deep Learning, ADHD at University of North Texas	Aug 2023 – May 2026
Competitive Programming Trainer — Chittagong University of Engineering & Technology	Jun 2017 – Jun 2020
Program Coordinator — CUET Computer Club	Sep 2020 – Dec 2020
Contributor — ABC of DataScience and ML Quora - https://abcofdatascienceandml.quora.com	Jun 2020 – Dec 2021
Member — IEEE	Jan 2018 – Present